

The roles of R&D team leaders in Korea: a contingent approach

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This study empirically examines the relationship between the role of R&D project leaders and their team performance using data from 87 project teams in 6 R&D organizations in Korea. The results reveal that:

- (1) R&D project leaders played five different roles in performing their jobs – strategic planner, team builder, gatekeeper, technical expert, and champion;**
- (2) All but the champion role of a leader is positively related with project team performance;**
- (3) However, this relationship between the role of leader and project team performance varies according to the characteristics of R&D project teams and their tasks.**

Specifically, it becomes important for a leader to focus less on the team building role as the team gets older. The team building role of a leader, however, is more important for higher performance of relatively certain R&D projects, while for uncertain R&D projects, the strategic planning role appears to be more crucial. Based on these results, this study discusses several managerial and theoretical issues related to the role of a leader in R&D project teams.

1. Introduction

Technological innovation is becoming a buzzword for managers and policy-makers in both advanced and developing countries. Indeed, many scholars and managers agree that R&D activities that foster technological innovation play a dominant role in acquiring and sustaining the competitive advantage of firms in an increasingly competitive global market. Recognizing the importance of effective management of R&D activities, many studies have explored diverse aspects of successful R&D projects such as project planning and evaluation, technical leadership, communication behaviour of R&D team members, and organizational structure and climate conducive to innovation (Allen, 1977; Farris, 1988; Katz and Tushman, 1981; Liberatore and Titus, 1983; O'Reilly, 1989; Pelz and Andrews, 1966; Tushman and Nadler, 1986).

Unfortunately, however, most of these studies have been undertaken in advanced countries, and few researchers focused on innovation activities in developing countries. Many firms in Newly Industrializing

Economies (NIEs), including Korea, are now facing a dilemma between technologically competitive forerunners from advanced countries and cost competitive followers from less developed countries, such as China and other south-east Asian countries. In response to their aspiration for a higher standard of living, the only step they can take is to catch up with advanced countries via accumulation of technological capability and development of competitive but high value-added products and services.

Given the limited resources for indigenous technology development activities, the effective management of R&D appears to be one of the most crucial strategic issues in these companies. Nevertheless, due to the differences in social culture, management practices, and the level of technological capability, it is not always relevant nor effective for these firms to imitate or follow R&D management systems developed in advanced countries. What is needed for them is to develop proper R&D management systems and practices, which fit local situations of NIEs. This can be accomplished by combining a careful observation of

local R&D using the inductive process and basic concepts and general frameworks derived from the deductive process (Kim and Kim, 1985).

This study, as an exploratory attempt to discover one important aspect of overall R&D management systems in a developing country, examines the relationships between the role of R&D project leaders and the project team performance based on empirical data from 87 R&D project teams selected from both public and private R&D organizations in Korea. In doing so, this paper first reviews existing literature on R&D project leadership and on critical functions needed in the innovation process for a derivation of important constructs and relationships among them. Then, it presents the research methods for sampling and measurement of variables, which is followed by the results of statistical analysis. Finally, it discusses several findings, which offer theoretical and managerial implications for effective R&D management in NIEs as well as in advanced countries.

2. Literature review

Undoubtedly, R&D performance depends substantially on technical leadership (Pelz and Andrews, 1966). Despite the greater number of studies on leadership in general, however, leadership in technological innovation has received relatively little systematic attention from academic scholars (Keller, 1992). Farris (1988) argues that technical leadership has been much discussed thus far, but it is not yet well understood. More systematic, replicable, and comparative studies of technical leadership are needed to determine which insights are specific to a particular situation and which are generalizable to many instances of technical leadership (Farris, 1988).

Theories of leadership outside R&D, though helpful in some ways in understanding technical leadership, operate in a different manner. For instance, conventional leadership styles have been found not to be able to fully explain the difference in research outcomes of R&D settings in existing studies (Andrews and Farris, 1967; Barnowe, 1975). Barnowe (1975) concluded that 'clearly, the better part of variation in research outcomes was attributable to factors other than leadership' (p. 273). The limited applicability of conventional leadership behaviour stems from the peculiar nature of both innovation task and R&D professional, which technical leaders must supervise in order to accomplish their missions.

Innovation tasks, contrary to other operating tasks, are intrinsically diverse, non-routine, and uncertain about the relationship between inputs and outputs, take a very long time for fruition, and usually involve a high risk of failure. Innovation processes are also plagued by many unforeseen disruptions, delays and setbacks (Quinn, 1979). To successfully manage an

innovation project, the technical leader, unlike a leader in charge of a routine job, must have a clear vision to cope with uncertain goals, creativity and persistence throughout the innovation process, an ability to cooperate and integrate team members with diverse backgrounds and, sometimes, even exploit serendipity or good luck. Successful innovations also require a lot of information and resources from outside of the R&D team, which must be addressed by the technical leader. However, conventional leadership behaviour has been usually confined to internal activities such as task structuring activities and human relationships with team members for task accomplishment and group maintenance (Ancona and Caldwell, 1988).

The R&D leader also deals with technical professionals who have many distinctive characteristics in terms of their goal orientation, value system, need structure, and their behavioural patterns (Badawy, 1988). They tend not to accept hierarchical control, unless they agree or understand, and frequently show less strong commitment to their organization than to their profession. They also tend to be chauvinistic about their technical specialty (Gomez-Mejia *et al.*, 1990; Miller, 1986; Von Glinow, 1988). Moreover, since autonomy is important in performing their work, the individual leader may have problems managing these professionals.

McCall (1988) suggested that effective technical leaders tend to affect technical performance of professional groups in two ways. They can take an active role in stimulating, evaluating and selecting various ideas and in setting work goals and objectives based on their technical competence. On the other hand, they also need to indirectly affect their professionals by nurturing and maintaining the work climate of creative tension, which seems to spawn high levels of professional productivity (Pelz and Andrews, 1966). For this, they must walk a tightrope between overcontrol and too much autonomy, make subtle changes in priorities and sequences, and ensure as well as channel professional challenges in the work (McCall, 1988).

Existing leadership research, however, does not provide a valuable insight into the functions and roles of R&D leaders. Recently, recognizing the limitations of earlier leadership studies, different approaches to technical leadership have been attempted. Farris (1988) classified existing studies of leadership in R&D settings into three categories: those which studied leadership directly, those which studied the characteristics of organizational climate which could be attributable to the leader, and those which studied the informal roles of R&D leaders who perform critical functions for project success.

The third approach, especially, has drawn much attention in recent studies, because it focuses on specific activities required for the leader in the innovation process and thus can offer substantial

managerial implications for the selection, training, role clarification or job description, and performance appraisal of the R&D project leader (Roberts and Fusfeld, 1981). These studies can be distinguished in terms of innovation context (i.e., R&D, new product development, technological change or innovation process), methodological approaches (i.e., conceptual works based on casual observation, case studies, or empirical studies), unit of analysis (i.e., individual, team or organizational unit) and scope of leaders' roles (i.e., specific role of champion or gatekeeper, or broad roles of the leader). Table 1 summarizes the characteristics of these studies.

In a nutshell, all these studies are based on the concept of critical functions in the innovation process. Many prior studies found that the innovation process typically encompasses several common critical functions performed by key individuals such as idea

generating, problem solving, information gatekeeping, project leading, championing, coaching, and team building functions (Allen, 1977; Pelz and Andrews, 1966; Souder, 1987). These functions are performed frequently by different R&D members, while some individuals occasionally play multiple roles (Roberts and Fusfeld, 1981). In particular, R&D project leaders are required to perform many diverse roles in order to manage successfully.

For instance, they often fulfil many functions in the innovation process such as coming up with original ideas, evaluating the critical ideas, helping to think about problems, providing a fair hearing for ideas, providing technical or organizational information to their team members, and interfacing with related external agencies including other functional groups (Farris, 1988). Table 1 shows the typical roles or functions of project leaders of innovation or new

Table 1 Summary of leader's roles and functions

Studies	Context	Empirical or conceptual	Level of analysis	Roles/Functions
Allen <i>et al.</i> (1988)	R&D project	Empirical	Team	Functional manager's role, project manager's role
Ancona and Caldwell (1988)	NPD* project	Case study	Team	Scout, ambassador, sentry, guard
Barczak and Wilemon (1989)	NPD project	Case study	Team	Communicator role, climate-setter role, planner role, interfacier role
Beatty and Lee (1992)	Technological change	Case study	Organization	Path-finding, problem solving, implementing
Chakrabarti and Hauschildt (1989)	Innovation process	Conceptual	Organization	Expert, champion, sponsor
Clark and Wheelwright (1992)	NPD project	Conceptual	Team	Direct market interpreter, multilingual translator, 'direct' engineering manager, program manager 'in motion', concept infuser
Farris (1988)	R&D	Conceptual	Individual	Captain: strategic leader, organizational leader, technical expert; Catalyst: informal leader, personal developer; climate creator, responsive leader
Friedman <i>et al.</i> (1992)	R&D	Empirical	Individual	Project management, personnel supervision, strategic planning
Howell and Higgins (1990)	Technology innovation	Empirical	Organization	Project champion, technical innovator, business innovator, chief executive, user champion
Maidique (1980)	Innovation process	Conceptual	Organization	Technologist, product champion, executive champion, sponsorship, technological entrepreneur
Markham <i>et al.</i> (1991)	Technology innovation	Empirical	Team	Champion, antagonist
McCall, Jr. (1988)	R&D	Conceptual	Team	Technical competence, controlled freedom, leader as metronome, work challenge,
Roberts and Fusfeld (1981)	Innovation process	Conceptual	Organization	Idea generating, championing, project leading, gatekeeping, sponsoring
Souder (1987)	Innovation process	Conceptual	Individual	Leader, capitalist, exciter, integrator, scout, linchpin, champion, mentor, translator, spotter

* New Product Development (NPD)

product development. From the table, five common roles could be delineated as representative roles a leader must fulfil to successfully complete the project. Three roles are concerned with the internal activities of the R&D team – technical expert, strategic planner, and team builder, while the other two roles are related to external activities – champion and information gatekeeper.

Technical expert. The role of technical expert encompasses many functions, including generating and recognizing good ideas, finding and defining the significant problems, providing technical stimulation, and fusing various ideas and solutions into an integrative framework or model. A consistent empirical result of R&D leadership shows that the technical expertise of R&D leaders have a direct and positive impact on technical performance (Farris, 1988). This is partly because professionals accept authority based on expertise rather than hierarchy, and partly because many critical functions in performing productive R&D activities depend on the technical expertise of the leader (McCall, 1988). Existing studies address the technical expert role of the leader by the name of ‘technical expert’ (Farris, 1988), ‘expert’ (Chakrabarti and Hauschildt, 1989), ‘idea generating’ (Roberts and Fusfeld, 1981), ‘technical innovator’ (Howell and Higgins, 1990), and ‘engineering manager’ (Clark and Wheelright, 1992).

Strategic planner. The conventional role of a project leader is to set the goals of the project, plan the time schedule and resource allocation, assign jobs to individual members, and control and evaluate the whole project process. However, R&D or innovation project leaders face difficulties in setting strategic goals and planning the means to achieve these goals, because innovation projects, by definition, attempt to conquer the unknown. Therefore the strategic planning role of a leader, which provides a strategic focus to the project and flexible planning to cope with many unforeseen contingencies during the innovation process (Barczak and Wilemon, 1989), appears to be critical for the success of innovation efforts. Many prior studies have revealed the importance of the strategic planner role of a leader by using various terms such as ‘strategic leader’ (Farris, 1988), ‘strategic planning’ (Fieldman *et al.*, 1992), ‘pathfinding’ (Beatty and Lee, 1992), ‘planner role’ (Barczak and Wilemon, 1989), ‘project leading’ (Roberts and Fusfeld, 1981), and ‘program manager in motion’ (Clark and Wheelright, 1992).

Team builder. The R&D leader by exerting technical leadership can positively affect team climate, which is conducive to individual creativity. Team cohesiveness or team spirit has been a necessary condition for building high performance project teams (Barczak and Wilemon, 1989). Indeed, team building is considered to

be one of the most critical leadership qualities that determine the performance of interdisciplinary or cross-functional efforts. To provide an atmosphere conducive to teamwork or transcendental effort (Nonaka, 1989), the project leader needs to develop team membership, resolve conflicts among members in a constructive way, and foster an environment where they have mutual trust and are willing to share different ideas, information, experiences and perspectives.

To instill a positive attitude among team members toward the project and to empower them, R&D leaders can resort to transformational leadership (Keller, 1992). They must articulate their vision and share it with team members to draw their commitment to accomplish it. They also stimulate the professional involvement of team members by emphasizing new challenges and innovative ways of doing things and encouraging risk-taking behaviour among members. They provide individual support by addressing the personal needs for growth and development of each team member (Bass, 1985). This role of a leader has been stressed in many studies by means of ‘climate creator’ (Farris, 1988), ‘climate setter’ (Barczak and Wilemon, 1989), and ‘exciter’ (Souder, 1987).

Champion. The task of innovation cannot be accomplished by only the aforementioned internal activities of the leader, because it must involve many interactions with external agencies for the supply of new information and other required resources. The role of a champion, who actively promotes innovation, builds support, gets required resources, overcomes resistance from organizational members, and ensures the implementation of the innovation (Howell and Higgins, 1990), has long been emphasized as a critical success factor of technical innovation (Chakrabarti and Hauschildt, 1989; Howell and Higgins, 1990; Roberts and Fusfeld, 1981; Schon, 1963).

Without the strong personal commitment and shrewd political tactics of project champions, technical innovations, especially radical innovations which require at least a sort of ‘creative destruction’ process, cannot be automatically accepted, nurtured, and implemented in the existing organization (Howell and Higgins, 1990; Katz, 1993; Kidder, 1981). There are always many different kinds of organizational resistance and reluctance to change or innovation, and the key ‘antagonists’ must be persuaded, convinced, or, sometimes even removed by the champion to overcome the barriers to get resources and support from related organizational units as well as from top management (Markham *et al.*, 1991). Ancona and Caldwell (1988) coined this role of technical leader ‘ambassador’, while many other studies called it ‘champion’ or ‘entrepreneur’ (Chakrabarti, 1974; Chakrabarti and Hauschildt, 1989; Roberts and Fusfeld, 1981; Souder, 1987).

Gatekeeper. The other frequently mentioned crucial function in the innovation process is technical gatekeeping (Allen, 1977; Katz and Tushman, 1981). Innovation process can be conceived as an information processing or creating process (Nonaka, 1989; Tushman and Nadler, 1986), and requires a great deal of market, organizational, and technical information for creative transformation. Many bits of information, however, need a mediating role for an effective communication: an identification of valuable sources, screening relevant information, and interpreting the implications of information. Gatekeepers, who occupy the key position in organizational communication networks, keep scanning and collecting information from external sources and disseminating the information with interpretation to the other internal organizational members (Allen, 1977).

The R&D project leaders have been found to play an important role in both internal and external communication (Allen *et al.*, 1988; Barczak and Wilemon, 1991; Katz and Tushman, 1981; Tushman and Scanlan, 1981). Many effective project leaders are identified as linking their teams with outside information sources, actively monitoring and receiving important information, relaying this information to members, and also facilitating communication among members. This can be accomplished through formal channels such as hierarchical reporting relationships, as well as informal linkages based on personal networks. Previous studies also acknowledge the gatekeeper role of the leader with 'communicator' (Barczak and Wilemon, 1989), 'direct market interpreter' or 'multilingual translator' (Clark and Wheelwright, 1992), 'scout' or 'guard' (Ancona and Caldwell, 1988).

Hypothesis 1: The five roles of an R&D leader mentioned above are positively associated with team performance.

Undoubtedly, some of these roles are more important in certain situations. Contingency theories of leadership have found that the effectiveness of the leadership is dependent on the situations the leader confronts, such as the task characteristics, nature of work groups and group members, and the past performance of the group (Jago, 1982). In R&D situations, many studies reveal that the leader's tenure with the project team, R&D team's age, and task uncertainty of the team can be important moderators of the relationship between the role of leader and project team performance (Allen *et al.*, 1988; Barczak and Wilemon, 1991; Keller, 1992).

Leader tenure. The effectiveness of a project leader can be influenced by the leader's tenure in the R&D team. Only when the leader has been with his team members long enough, can the leader make a substantial impact and influence the behaviour of team members and team climate, which in turn affects

project performance. Because it takes a relatively long time for the technical leader to gain the confidence of his R&D professionals who are used to autonomy and not easily influenced by formal authority, it is never easy for the leader to instill his goals and values into his team members and to change behavioural patterns established during the long period of professional education. Even after they have accepted the influence of the leader, it would take time to bear fruit.

Hypothesis 2: The relationship between the R&D leader's role and team performance is moderated and reinforced by leader's tenure

Team tenure. NIH or Not-Invented-Here syndrome means 'the tendency of a project group of stable composition to believe it possesses a monopoly of knowledge of its field, which leads it to reject new ideas from outsiders to the likely detriment of its performance' (Katz and Allen, 1982, p. 7). As project teams become increasingly cohesive and stable enough to feel complacent over time, they begin to separate or isolate themselves from external sources of technical ideas and influence, and communicate less frequently with outside professional groups. NIH syndrome stems from the basic tendency that over time individuals try to organize their work environment in a manner that reduces the amount of stress and uncertainty they must face. This individual tendency is reinforced by the situation where all the project members work together and share similar experiences for a certain amount of time (Katz, 1981). They are more likely to rely on their existing knowledge, views, experiences, and capabilities for performing their tasks than to resort to outside information and expertise. They become increasingly immune to outside technology development trends.

Given the importance of discussing and transferring technical information and new ideas into the innovation process, it is not surprising that Katz and Allen (1982) found the performance of R&D project teams declining noticeably after a team's tenure of five years, together with the communication level among team members and outside information sources. This result implies that as the team tenure prolongs, it becomes more crucial for the project leader to make a deliberate effort to avoid NIH syndrome. The leaders must help their team neither to become complacent nor be isolated from outside information sources. They should keep challenging team members, facilitating communication, encouraging new approaches, and linking team members with critical external agencies. On the contrary, since the team building role of a leader intends to foster the cohesiveness and consolidation of team members, it is more likely to reinforce NIH syndrome and have a negative impact on the performance of a long-tenured team. In other words, the gatekeeper and champion roles of a project leader become more crucial for enhancing team performance,

while the role of team builder may have a negative effect on the R&D performance of the project team.

Hypothesis 3: As the team tenure increases, the leader's roles of gatekeeper and champion are positively associated with the team performance, while the role of team building is negatively associated with the team performance.

Task characteristics. The role of leaders must also be contingent to the type of R&D project. Keller (1992) found that different leadership styles are required for different projects. For instance, research projects are more concerned with innovation and the exploitation of scientific and technological information from outside, while development projects, which generally focus on the improvement of existing technology, require the leader to focus on task allocation and coordination skills. Indeed, the project leader must satisfy the different requirements of different R&D tasks with different role emphasis.

When the task uncertainty of the R&D team is high, prior research has found that both internal and external communication activities must also be high and that project leaders must share their communication activity and gatekeeper role with other team members for better team performance (Katz and Tushman, 1981; Tushman, 1979; Tushman and Scanlan, 1981). In an uncertain task environment, high performing project teams are more likely to rely on a small number of boundary spanning individuals for external communication activities, who are not necessarily the team leaders. However, this does not mean that the gatekeeper or communication role of a project leader becomes weaker, as the R&D project task becomes more uncertain. Rather, project leaders must exhibit aggressive behaviour to process and create more information together with help from other team members, as they have to surmount highly complex and dynamic technical situations in order to accomplish their innovation projects.

When the R&D project faces greater uncertainty in market and/or technical environments, it becomes more critical for the reduction of uncertainty or confusion among team members to have clear strategic goals or targets for the project, which used to be very difficult to formulate under uncertain circumstances. On the other hand, the team building role of a leader can be viewed as more important when the team faces a more clearly defined task environment, because it is more important to have coherent approach or good team cohesion for the efficient operation of routine R&D tasks.

Hypothesis 4: When the task uncertainty of the team is high, the leader's roles of gatekeeper and strategic planner are more positively associated with team performance. However, when the task uncertainty is

low, the leader's role of team builder is more positively associated with team performance.

3. Research methods

Samples

One hundred and three project teams were sampled from three government-sponsored and three private R&D organizations, which were selected in a pair-wise manner in the areas of electronics, chemistry, and mechanical engineering. *In toto*, questionnaires were collected from 570 R&D professionals in 103 project teams. However, only 87 project teams, comprising 503 professionals, were finally used for further data analysis, after excluding the teams which responded to less than four questionnaires.

Measurements

Roles of leader. A multi-item scale was developed to measure the role of R&D project leader based on the literature review and pilot interviews with several R&D project leaders in both government-sponsored and private research institutes. The initial items were extracted from various studies of the critical functions of innovation and of the roles of R&D leader (see Table 1), and then were verified for their relevance to the local R&D situation through structured interviews with R&D project leaders. About half of the team members, excluding the leader (on average 2.6 per team) were randomly chosen to rate the scale of leader roles and the individual scores were aggregated at the team level.

As shown in Table 2, five factors are selected on an *a priori* basis from the principal component analysis, which account for 76% of total variance. There are many criteria for the extraction of factors, including eigenvalue, scree-test, *a priori* criteria, and so on (Hair *et al.*, 1979). Since this study conceptualized the five different roles of the leader, the *a priori* criteria have been adopted. Each one of the five factors quite well represents the concept of five different roles of the R&D leader, although two of the factors have relatively low eigenvalues and thus result in low explanation powers. All of the factor scores used in subsequent analyses were created by assigning factor loading weights to item responses¹. Table 3 also reveals satisfactory results of the reliability test of the measurement of the leader role. All the Cronbach alpha coefficients of leader role variables exceed 0.78, which are very acceptable according to a suggested guideline by Van de Ven and Ferry (1980).

In addition, in order to justify the aggregation of individual scores of leader roles at the team level, this study followed the James *et al.* (1984) procedure for calculating within-group inter-rater reliability. This

procedure is based on a comparison of observed within-group agreement to the agreement one would expect by chance in the group. Table 2 indicates that the averages of *rwg* (*j*) are 0.77, 0.75, 0.74, 0.76, and 0.74 for technical expert, team builder, champion, gatekeeper, and strategic planner factor of the leader role, respectively. These agreement estimates are considered adequate² because the James *et al.* (1984) formula usually underestimates inter-rater agreement for groups less than 10 respondents.

Team performance. R&D team performance was measured by the scale composed of five items, comprising achievement of technical objectives, efficiency of resource utilization, staying on schedule, value to the organization, and overall goal achievement beyond expectation. The five performance criteria used in this study are quite similar to those in other studies (Ancona and Caldwell, 1990; Keller, 1994). R&D team performance was measured by an aggregate score of all the members of the team, including the team leader. The average number of respondents for team performance measurement is 5.7³.

A principal component analysis produced only one factor with the eigenvalue greater than one. The

Cronbach alpha coefficient of team performance was 0.73 and inter-rater agreement coefficient, *rwg* (*j*), resulted in 0.92.

Leader tenure. It was measured by the number of months the leader was associated with a particular R&D team or a group.

Team tenure. It was computed based on the average tenure of individual team members with the R&D team or group.

Task uncertainty. An instrument developed by Withey *et al.* (1983) was used in this study. This instrument produced two dimensions of task uncertainty: task analysability and task variability (Perrow, 1967). Cronbach alpha coefficients of task analysability and task variability are 0.91 and 0.75, respectively. Project teams which have higher scores than the median in both dimensions were classified into high uncertain group (*n* = 25), while those with lower scores in both dimensions were classified into low uncertain group (*n* = 26) (Dunegan *et al.*, 1992). Other project teams (*n* = 36) were excluded in the analysis of task uncertainty.

Table 2. The results of factor analysis on the leader's role (*N* = 87)

Leader's role item summary	<i>Rwg</i> ^a	Factor loadings				
		Factor I	Factor II	Factor III	Factor IV	Factor V
<i>Technical expert</i>	0.77					
1. Suggest problem solution	0.74	0.85	0.20	0.14	0.13	0.19
2. Generate new idea	0.78	0.83	0.24	0.17	0.14	0.19
3. Appraise possibility/problem	0.75	0.61	0.40	0.26	0.20	0.17
4. Technical consultant	0.75	0.61	0.21	0.38	0.28	0.20
<i>Team builder</i>	0.75					
5. Conflict solving/integrator	0.76	0.20	0.82	0.12	0.14	0.21
6. Cohesiveness builder	0.73	0.24	0.78	0.22	0.22	0.11
7. Team climate setter	0.74	0.21	0.78	0.27	0.07	0.18
8. Considering and empowerment	0.70	0.29	0.70	0.21	0.30	0.18
<i>Champion</i>	0.74					
9. Resistance/risk taking	0.77	0.35	0.34	0.75	0.01	0.01
10. Give product (production) form	0.79	0.39	0.15	0.71	0.05	0.28
11. Acquire supports and resources	0.70	0.11	0.22	0.70	0.49	0.10
12. Coalition building for support	0.76	0.02	0.31	0.58	0.47	0.33
<i>Gatekeeper</i>	0.76					
13. Contact with information sources	0.73	0.30	0.21	0.14	0.76	0.22
14. Diffuse technical/market information	0.76	0.55	0.24	0.14	0.59	0.11
<i>Strategic planner</i>	0.74					
15. Planning/scheduling	0.77	0.31	0.16	0.10	0.09	0.83
16. Task coordination	0.74	0.09	0.34	0.27	0.40	0.59
17. Strategic goal setting	0.77	0.43	0.35	0.20	0.28	0.53
Eigenvalue		8.98	1.27	1.09	0.95	0.71
Cumulative %		0.53	0.61	0.67	0.72	0.76

(a) James *et al.* (1984)'s *Rwg* is the within group interrater reliability statistics indicating the level of agreement among judge's ratings of a single target. See James *et al.* (1984) for computational formula.

Table 3. Number, mean, standard deviation, Pearson correlation and reliability (Cronbach α)

	N	M	S.D.	1	2	3	4	5	6	7	8	9	10
<i>Leader's Role</i>													
1. Technical expert	87	-0.02	0.82	(0.90)									
2. Team builder	87	0.04	0.75	0.09	(0.89)								
3. Champion	87	0.01	0.87	0.06	0.05	(0.85)							
4. Gatekeeper	87	-0.03	0.79	0.06	0.10	0.02	(0.79)						
5. Strategic planner	87	-0.01	0.76	-0.04	0.02	-0.01	0.12	(0.78)					
<i>Contingency variables</i>													
6. Leader tenure (months)	74	29.2	27.2	0.05	-0.04	0.06	-0.10	-0.12	(—)				
7. Team tenure (months)	75	56.7	13.8	-0.13	-0.14	-0.17	-0.04	-0.07	-0.14	(—)			
8. Variety	86	3.45	0.53	0.22***	0.22***	0.36***	0.04	0.13	-0.05	-0.17*	(0.91)		
9. Analysability	86	2.92	0.35	0.03	0.08	-0.06	0.26*	0.01	0.00	0.21*	0.45***	(0.75)	
<i>Performance</i>													
10. Team performance	87	3.40	0.34	0.35***	0.27***	0.06	0.39***	0.25**	-0.28**	0.05	0.23**	0.13	(0.73)

*, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$

4. Results

Table 3 presents means, standard deviations, and a correlation matrix for the variables.

All but the champion role of a leader is significantly associated with team performance. Further, multiple regression analysis was performed to see whether these relationships are borne out in a multivariate context. The results of regression analysis in Table 4 confirm hypothesis 1, except for the champion role of a leader. The results also indicate that the four roles of the leader account for 35% of the total variance of R&D team performance.

One may argue, however, technical expert represents 53% out of 76% of total variance in the principal component analysis, as Table 2 shows, and thus appears an utmost important role for the R&D leader with other roles of minor importance. This is not surprising in the context of R&D activities, since technical innovation itself requires scientific and technical knowledge and deep understanding of the technical problems to be solved. Moreover, R&D people themselves are technical professionals, who tend to respect a leader who has excellent technical expertise, rather than hierarchical authority (Miller, 1986).

Existing studies in advanced contexts already recognized the importance of technical expertise for R&D leadership. For instance, Farris (1988, p. 14) asserted that 'First and foremost is the importance of the leader's technical skills. Much more consistently and strongly than in studies of non-R&D situations, the technical skills of technical supervisors are related to the performance of their groups, and especially to innovation by the groups.' McCall (1988) also emphasizes the importance of technical skill in implementing

R&D leadership, since many roles R&D leaders must perform depend on technical expertise, including recognizing and communicating good ideas (*gatekeeper*), influencing work goals (*strategic planner*), persuading other stakeholders inside and outside the organization (*championing*).

This does not mean, however, that other roles the R&D leader must perform are not important for the R&D team performance. Table 4 indicates that leader roles other than technical expert account for a significant portion of R&D team performance. Thus it would be fair to conclude that while technical expertise of an R&D leader itself is very crucial for the team performance, other roles must also be fulfilled by the leader for the performance of R&D projects, although many may not be successfully performed without technical expertise.

Table 5 shows the results of moderated regression analyses to examine the interaction effects of leader tenure on the relationship between the leader's roles and team performance. As predicted in hypothesis 2, when the leader's tenure extends, the relationship between the roles of leaders and R&D team performance was found to be stronger except for the

Table 4. Regression results regressing team performance on leader's five roles ($n = 86$)

Leader's role	β Coefficient	R^2	F-value
Technical expert	0.32***	0.35	8.516***
Team builder	0.20**		
Champion	0.33		
Gatekeeper	0.33***		
Strategic planner	0.22**		

*, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$

Table 5. Regression results regressing team performance on leader tenure, leader's role, and the interaction ($n = 74$)

Step	Variable	β Coefficient	R^2	ΔR^2	F-value (ΔR^2)
1.	Leader tenure	-0.004	0.078	0.078	6.09
2.	Technical expert	0.191***	0.264	0.186	18.448***
3.	Leader tenure* expert	0.003*	0.302	0.038	3.974*
2.	Team builder	0.123**	0.141	0.063	5.354**
3.	Leader tenure* builder	0.003	0.159	0.018	1.562
2.	Champion	0.043	0.087	0.009	0.720
3.	Leader tenure* champion	0.000	0.087	0.000	0.000
2.	Gatekeeper	0.186***	0.251	0.173	16.861***
3.	Leader tenure* gatekeeper	0.004**	0.303	0.052	5.446**
2.	Strategic planner	0.117**	0.139	0.061	5.171**
3.	Leader tenure* planner	0.006***	0.221	0.082	7.684***

*, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$

Table 6. Regression results regressing team performance on team tenure, leader's role, and the interaction ($n = 75$)

Step	Variable	β Coefficient	R^2	ΔR^2	F-value (ΔR^2)
1.	Team tenure	0.001	0.002	0.002	0.17
2.	Technical expert	0.168***	0.157	0.155	13.606***
3.	Team tenure* expert	-0.006	0.186	0.029	2.636
2.	Team builder	0.105*	0.045	0.043	3.332*
3.	Team tenure* builder	-0.013***	0.164	0.119	10.533***
2.	Champion	0.016	0.003	0.001	0.074
3.	Team tenure* champion	0.000	0.003	0.000	0.000
2.	Gatekeeper	0.166***	0.147	0.145	12.579***
3.	Team tenure* gatekeeper	-0.004	0.155	0.008	0.701
2.	Strategic planner	0.129**	0.078	0.076	6.100**
3.	Team tenure* planner	-0.003	0.088	0.010	0.811

*, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$

champion and team building roles. When the leaders have enough time to interact with team members, they have more influence on team members' activities and in turn on team performance.

Table 6 presents the results of moderated regression analyses to test the interaction effects of team's tenure on the relationship between the leader roles and team performance. The result shows that there is a negative impact of the team builder role of a leader on team performance for longer-tenured teams. However, the positive interaction effects of gatekeeper and champion

roles of a leader on team performance, as hypothesis 3 predicts, are not borne out.

Table 7 presents the moderating effects of task characteristics on the relationship between the leader's role and team performance. The results reveal that when the task uncertainty of an R&D team is relatively low, the team building role of a leader has a significant positive relationship with team performance. As the R&D team task becomes more uncertain, on the contrary, the role as a strategic planner is more likely to be important to R&D team performance. The roles as an expert and a gatekeeper appear to be crucial for the R&D team performance, no matter how uncertain the R&D task of the team is.

Table 7. Correlation between leader's roles and team performance for task uncertainty and Fisher's Z test

Leader's role	Task uncertainty		Fisher's Z test
	Low ($n = 26$)	High ($n = 25$)	
Technical expert	0.31	0.42**	-0.44
Team builder	0.43**	-0.01	1.58*
Champion	-0.14	0.17	-1.04
Gatekeeper	0.39**	0.33*	0.23
Strategic planner	-0.14	0.52***	-2.41***

*, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$

5. Summary and discussion

In an effort to produce a suitable R&D management system in a developing context, this study empirically explores the effective roles of R&D team leaders on the basis of 87 R&D project teams in both the public and private sectors in Korea. The results show that all but the champion role of a leader among the five required roles – technical expert, strategic planner, team

builder, and gatekeeper – are positively related with team performance. These relationships are more evident, if the leader has worked together with team members for a long period of time, long enough for his efforts to have impact. This study also found that team tenure and task characteristics moderated the relationship between leader's roles and team performance. For instance, the leader of longer-tenured teams must focus less on the team builder role in order to avoid possible NIH syndrome. The team builder role of a leader, however, tends to be more important for the success of a relatively well-defined R&D project, while, for ill-defined projects, the strategic planner role appears to be more crucial for the team performance.

These findings have both commonality and differences compared to those in existing studies. First, the required roles of an R&D leader in Korea do not seem to be much different from those found in advanced countries. Five roles of an R&D leader derived from the factor analysis in the study have commonality with those found empirically in various innovation contexts (see Table 1). This might stem from the fact that the innovation process basically requires several critical functions to be accomplished in any context (Roberts and Fusfeld, 1981).

Second, the champion role of a leader, however, does not appear to be relevant to the success of R&D projects in Korea. In advanced countries, there are many successful heroic stories of innovation championship (Katz, 1993; Kidder, 1981; Nayak and Ketteringham, 1994) and recently many studies have been undertaken to reveal the nature of the champion role in the innovation process (Howell and Higgins, 1990; Markham *et al.*, 1991; Shane, 1994, 1997). There are a couple of possible reasons to explain the insignificance of the champion role of the R&D leader in Korea.

Shane (1998) suggested that the more individualism and the less power distance and uncertainty avoidance are valued in a society or organization, the more prevalent championing activities become. The cultural characteristics in Korea, however, can be characterized by lower individualism, relatively higher power distance and uncertainty avoidance, compared to the USA and other advanced western cultures (Hofstede, 1992). Given these cultures, it is not surprising that most R&D projects in Korea have been planned and selected from the top management of organizations. It is hardly conceivable for the project leader to initiate the R&D project without *a priori* formal consent of top managers. In addition, many R&D projects in this study are more likely to be imitative and incremental in nature and focus less on the commercialization of new technologies. Performing these projects may not involve high risk of failure, and neither need strong task coordination with other organizational units nor, in turn, require the champion role of an R&D leader.

These results offer an implication for the development of R&D management theories in a developing context. There are basically two ways to extend existing knowledge and to develop a new theory: formulating an explanation for the observation of real world facts using the inductive approach and generating predications about the phenomena based on deductive approaches (Stone, 1978). Research on the management of innovation in a development context can benefit from knowledge generated in advanced contexts for a general conceptual framework and at the same time generate significantly more creative and relevant hypotheses when the local situation is taken into consideration in the research design (Kim and Kim, 1985). By combining both the deductive and inductive approaches in an interactive manner, the R&D management system, which is relevant to the local situation in developing countries, can be efficiently constructed.

This study, based on a conceptual framework developed in advanced countries, found both similar and different relationships between the role of an R&D leader and team performance, compared with those found in existing studies. Similarity may represent fundamental requirements of innovation management, while dissimilarities can reflect different local R&D situations with respect to the level of technological capability, nature of R&D project, social culture, and organizational structure (Kim and Lee, 1995; Kim and Utterback, 1983).

This study also suggests several managerial implications for R&D leadership and human resource management in Korea. The R&D project leaders themselves must be aware of the different role requirements of the situations. First, the leaders must change their relative role emphasis as their team evolves along time. In the initial stage, they must focus on strategic planner and team builder roles to provide a clear common understanding of the project missions or goals and to establish cohesiveness among team members. Yet, they must be patient to see any performance feedback of their leadership activities, because it usually takes a considerable time for their efforts to show measurable effects. However, when their teams have been together for long enough to suffer from NIH syndrome, they should de-emphasize the role of team building and emphasize more on interfacing with external agencies and stimulating or challenging team members to prevent the tendency towards isolation from the environment and complacency with the existing programmed activities.

Second, the leaders must also consider the type of projects they perform. Fundamental or radical R&D projects basically involve more uncertainty in goal setting and require more activity in processing and creating scientific, technical, and market knowledge. Thus they must make more efforts to set clear strategic

goals and to design flexible plans to respond to unforeseen contingencies or serendipity. They should also carry out the technical expert role by suggesting new ideas and alternative technical solutions for themselves. However, when they are in charge of relatively routine, incremental development tasks, activities for consolidation and empowerment of team members must be more emphasized to a greater extent (Tushman, 1979).

Third, our findings suggest that the diagnostic activities of the leader role can be used for many purposes such as selection, performance appraisal, staffing, and design of training programmes and career development paths. A suitable R&D project leader can be selected and evaluated on the basis of a careful appraisal of the potential leader's qualification to perform various roles, in conjunction with the characteristics of the tasks and teams he or she has to supervise. The results also are helpful in developing different skill training for leaders with different role performance profiles (i.e., communication, interpersonal, team-building, problem-solving, or technical skills), and to decide a viable career path for them (research specialist, R&D manager, or transfer to the business division).

Finally, this study is basically cross-sectional in nature. It surely warrants a longitudinal approach to conclude the causal relationship between the role of leaders and their team performance. It is also necessary for further studies to refine the measurement of leader's roles and R&D team performance. Among the five factors of the roles of R&D leader are only three factors whose eigenvalues exceed one, as Table 2 indicates. To improve discriminant and convergent validities of the measurement, the multi-item scale of the leader's role can be refined based on the results of the current study. Another measurement approach can be applied for different leader roles. For instance, sociometric methods can be employed for the measurement of the role of a gatekeeper (Allen, 1977; Tushman and Scanlan, 1981), or more qualitative methods such as content analysis can be used for the champion role (Howell and Higgins, 1990) and for other leadership roles (Barczak and Wilemon, 1989). In addition, objective indicators for team performance should be developed in future studies.

The results of this study should also be extended to other innovation contexts. Projects for the commercialization of new products or processes may require a slightly different leadership roles for successful implementation, and innovation projects aimed at the adoption of new administration systems or information technologies in the organization might also need different leadership profiles. Future studies can address these differences in diverse innovation contexts with more refined measurements of the role of technical leaders.

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Notes

1. The five roles are very much correlated among themselves, if arithmetic means of each role variable are used. To avoid the multi-collinearity problem, thus, factor scores were used for leader roles in the subsequent analyses.
2. The *rwg(j)* for Multifactor Leadership Questionnaire was calculated by Hater and Bass (1988). That ranges from 0.67 to 0.90 and the mean is 0.79. (In this study, the mean is 0.75).
3. One may raise the issue of common method bias in measuring team performance and leader roles in the study. However, unit of analysis in this study is the team, not individual, and the number of respondents for each measure is different (2.6 for leader role and 5.7 for team

performance). Furthermore, we undertook a principal component analysis with both variables for Harman's single-factor test. If a substantial amount of common method variance is present, there will emerge a single factor from the factor analysis, or one general factor will account for the majority of the covariance in the independent and criterion variables (Podsakoff and Organ, 1986). As a result

of factor analysis, six factors derived from unrotated factor structure, which represent five different leader roles and a team performance, respectively. The result indicates that the relationships between leader roles and team performance were not merely caused by a measurement error, common method bias.